In the name of God

Department of Electrical Engineering

Software-Defined Radio Laboratory
Lab 2 Pre-lab Report

Noise Figure and RF Parameters

Group 6

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1 Summary of concepts

The noise factor shows how much the signal-to-noise ratio falls as a signal passes through a receiver. If the source is at the reference temperature $T_0 = 290\,K$, we have

$$F = \frac{SNR_{in}}{SNR_{out}} = 1 + \frac{T_e}{T_0}, \qquad NF = 10\log_{10} F$$

where T_e is the equivalent noise temperature of the receiver, referred to its input. So the output noise is

$$N_{out} = GkB(T_{in} + T_e)$$

In a chain, the noise added by each stage is divided by the gain before it. So if the first stage has low noise and enough gain, the later stages have less effect. Loss before the first stage directly increases the total noise figure.

To measure the noise figure with the Y-Factor method, we read the output noise power with the source on and off. If the gain and filter stay the same, the gain, bandwidth, and power conversion factor cancel in the ratio. So we do not need an absolute power measurement. With the off-state temperature equal to T_0, we have

$$Y = \frac{N_{on}}{N_{off}}, \qquad F = \frac{ENR}{Y - 1}$$

The receiver must stay out of compression with the source on, and automatic gain control must be off. Also, if Y is too close to one, a small error in the ratio can cause a large error in the noise figure.

To study nonlinearity, we apply two tones close in frequency. The third-order term creates components at $2f_1 - f_2$ and $2f_2 - f_1$. In the weakly nonlinear region, their power rises by about three decibels for each decibel of input increase. If we extend the fundamental and product lines, their intersection gives $IIP3$. This value predicts the growth of the third-order products. It does not mean that the receiver still works linearly at that input level.

Increasing the gain reduces the input-referred ADC noise, but it also makes the signal reach the ADC limit earlier. So we need to consider both the noise floor and the ability to handle strong signals when we choose the gain.

The range from the noise floor to the largest signal whose products remain below the noise is the spur-free dynamic range. In the valid third-order region, we have

$$N_{floor} = -174 + NF + 10\log_{10} B$$
$$SFDR = \frac{2}{3}(IIP3 - N_{floor})$$

2 Theory questions

2.1

From the equivalent noise temperature relation, we know that

$$T_e = T_0(F-1) = 290(10^{NF/10}-1)$$

Substituting the four given noise figures gives the following table.

$NF\ (dB)$	F	$T_e\ (K)$
1	1.258925	75.088
3	1.995262	288.626
6	3.981072	864.511
10	10.000000	2610.000

To calculate the noise figure of the chain, we first convert all gains and noise factors to linear values. We assume that the filter is matched and at T_0. Then

$$\begin{aligned} G_1 &= 10^{20/10} = 100, \qquad F_1 = 10^{1/10} = 1.258925 \\ L &= 10^{3/10} = 1.995262, \qquad G_2 = \frac{1}{L} = 0.501187 \\ F_2 &= L, \qquad F_3 = 10^{15/10} = 31.622777 \end{aligned}$$

The filter noise factor equals its loss under this temperature assumption. Now we use the Friis relation:

$$\begin{aligned} F &= F_1 + \frac{F_2-1}{G_1} + \frac{F_3-1}{G_1G_2} \\ &= 1.258925 + \frac{1.995262-1}{100} + \frac{31.622777-1}{100\times 0.501187} \\ &= 1.879883 \end{aligned}$$

So the result is

$$\boxed{NF = 10\log_{10}(1.879883) = 2.741308\,dB}$$

If we put the filter before the amplifier, the order of the stages changes. This time,

$$\begin{aligned} F' &= L + \frac{F_1-1}{1/L} + \frac{F_3-1}{(1/L)G_1} \\ &= 3.122891 \end{aligned}$$

and therefore

$$\boxed{NF' = 4.945568\,dB}$$

This is about $2.204261\,dB$ higher than before. The reason is clear: the filter now reduces the signal before it is amplified. In the first arrangement, the filter noise contribution was divided by the amplifier gain of 100.

2.2

First, we call the on-state temperature T_h and the off-state temperature T_c. From the output noise relation,

$$N_h = GkB(T_h + T_e), \qquad N_c = GkB(T_c + T_e)$$

We divide these two equations:

$$Y = \frac{T_h + T_e}{T_c + T_e} \quad \Rightarrow \quad T_e = \frac{T_h - YT_c}{Y - 1}$$

If $T_c = T_0$, using the definition $ENR = (T_h - T_0)/T_0$, we can write

$$\begin{aligned} Y - 1 &= \frac{T_h - T_0}{T_0 + T_e} \\ &= \frac{(T_h - T_0)/T_0}{1 + T_e/T_0} = \frac{ENR}{F} \end{aligned}$$

So we obtain the required relation:

$$\boxed{F = \frac{ENR}{Y - 1}}$$

If the off-state temperature is different from the reference temperature, we use the general relation:

$$F = 1 + \frac{T_h - YT_c}{T_0(Y - 1)}$$

Keeping the handout's definition of ENR and substituting $T_h = T_0(1 + ENR)$ gives

$$\boxed{F = \frac{ENR + Y(1 - T_c/T_0)}{Y - 1}}$$

To study the attenuator, we call its linear loss L and its loss in decibels $A = 10 \log_{10} L$. Its physical temperature is also T_0. For a source at temperature T, the output noise temperature is

$$T' = \frac{T}{L} + \left(1 - \frac{1}{L}\right) T_0$$

In the off state, we still have $T_c' = T_0$. For the on state,

$$T_h' - T_0 = \frac{T_h - T_0}{L}$$

As a result,

$$ENR_{eff} = \frac{ENR}{L} \quad \Rightarrow \quad \boxed{ENR_{eff,dB} = ENR_{dB} - A}$$

So the effective ENR falls by exactly the attenuator loss in decibels.

2.3

To calculate the sensitivity, we define the ratio in decibels as $y = 10 \log_{10} Y$. Then

$$NF = ENR_{dB} - 10 \log_{10}(10^{y/10} - 1)$$

Taking the derivative with respect to y gives

$$\frac{\partial NF}{\partial y} = -\frac{10^{y/10}}{10^{y/10} - 1} = -\frac{Y}{Y - 1}$$

In the first case, with $NF = 6\,dB$ and $ENR = 15\,dB$, we first calculate the true ratio:

$$Y = 1 + 10^{(15-6)/10} = 8.943282, \qquad y = 9.514969\,dB$$

The absolute value of the sensitivity coefficient is 1.125893. For an error of $0.1\,dB$ in the ratio reading, the first-order estimate of the noise figure error is $0.112589\,dB$. Substituting the changed ratio into the full equation gives the exact errors in the table below.

In the second case, with $NF = 25\,dB$, we have

$$Y = 1 + 10^{(15-25)/10} = 1.1, \qquad y = 0.413927\,dB$$

The sensitivity coefficient is now 11. So the estimated error becomes $1.1\,dB$.

NF (dB)	y (dB)	NF error for $y + 0.1$	NF error for $y - 0.1$
6	9.514969	-0.112428	0.112754
25	0.413927	-0.990667	1.251650

In the second case, the exact positive and negative errors are not equal because the relation is nonlinear in y. The sensitivity also increases as Y gets closer to one. So when choosing ENR, the on and off readings should differ by several decibels. At the same time, the source must not be so strong that it causes compression. For a starting point in this experiment, an $ENR - NF$ difference of about 5 to 10 decibels is suitable. We then check that the receiver stays linear.

2.4

We consider two input tones with equal amplitudes:

$$x(t) = A \cos \omega_1 t + A \cos \omega_2 t, \qquad y = a_1 x + a_3 x^3$$

To find the products, we expand the cube of the input. If $c_1 = \cos \omega_1 t$ and $c_2 = \cos \omega_2 t$, we have

$$x^3 = A^3(c_1^3 + 3c_1^2 c_2 + 3c_1 c_2^2 + c_2^3)$$

Now we use these two identities:

$$\cos^3\alpha = \frac{3}{4}\cos\alpha + \frac{1}{4}\cos 3\alpha$$
$$\cos^2\alpha\cos\beta = \frac{1}{2}\cos\beta + \frac{1}{4}\cos(2\alpha+\beta) + \frac{1}{4}\cos(2\alpha-\beta)$$

The amplitudes of the fundamentals and the nearby products are then

$$B_{f_1} = B_{f_2} = a_1 A + \frac{9}{4}a_3 A^3$$
$$B_{2f_1-f_2} = B_{2f_2-f_1} = \frac{3}{4}a_3 A^3$$

In the small-signal region, the first term of the fundamental is dominant, so its amplitude is proportional to A. The product amplitude is proportional to A^3. Since the input power per tone is proportional to A^2, we have

$$P_{fund} \propto P_{in}, \qquad P_{IM3} \propto A^6 \propto P_{in}^3$$

This gives a slope of one for the fundamental and three for the products on a decibel scale.

Suppose the gap between the fundamental and product at one input level is Δ. When the input increases by δ decibels, the gap falls by 2δ. So the extended lines meet when $\delta = \Delta/2$. Therefore,

$$\boxed{IIP3 = P_{in,per\,tone} + \frac{\Delta}{2}}, \qquad OIP3 = IIP3 + G$$

The input power in this equation is the power of each tone.

To derive the compression relation, we first find the intercept amplitude by setting the two extended amplitudes equal:

$$|a_1|A_{IP3} = \frac{3}{4}|a_3|A_{IP3}^3 \quad \Rightarrow \quad A_{IP3}^2 = \frac{4|a_1|}{3|a_3|}$$

For a single-tone input, the fundamental amplitude is $a_1 A + \frac{3}{4}a_3 A^3$. For compression, we assume $a_1 > 0$ and $a_3 < 0$. At the one-decibel compression point,

$$1 - \frac{3|a_3|}{4a_1}A_{1dB}^2 = 10^{-1/20}$$

Using the intercept amplitude, we can write this as

$$A_{1dB}^2 = A_{IP3}^2(1 - 10^{-1/20})$$

Since power is proportional to the square of amplitude,

$$\begin{aligned} IIP3 - P_{1dB,in} &= -10\log_{10}(1 - 10^{-1/20}) \\ &= 9.635745\,dB \end{aligned}$$

So

$$\boxed{IIP3 \simeq P_{1dB,in} + 9.6\,dB}$$

There are two limits to this result. First, a memoryless third-order model must describe the stage well. Higher-order terms and memory effects can change the gap. Second, this nonlinearity must cause the compression. If ADC clipping or automatic gain control acts first, this relation no longer applies.

2.5

To calculate $SFDR$, we first find the noise floor in each bandwidth. From the handout,

$$N_{floor} = -174 + NF + 10\log_{10} B$$

For $NF = 8\,dB$ and $IIP3 = -10\,dBm$, we can write

$$\begin{aligned} B = 200\,kHz: &\quad N_{floor} = -112.989700\,dBm \\ B = 10\,kHz: &\quad N_{floor} = -126\,dBm \end{aligned}$$

To explain the factor of two-thirds, we refer the product power to the input. At the point where the product reaches the noise floor,

$$3P_{max} - 2IIP3 = N_{floor}$$

So

$$\begin{aligned} SFDR &= P_{max} - N_{floor} \\ &= \frac{N_{floor} + 2IIP3}{3} - N_{floor} \\ &= \frac{2}{3}(IIP3 - N_{floor}) \end{aligned}$$

Substituting the two noise floors gives

$$\boxed{SFDR_{200kHz} = 68.659800\,dB, \qquad SFDR_{10kHz} = 77.333333\,dB}$$

The bandwidth matters because noise power is added across the whole band. In this example, increasing the bandwidth by a factor of 20 raises the noise by $10\log_{10} 20 = 13.010300\,dB$. As a result, $SFDR$ falls by $8.673533\,dB$.

2.6

For the 8-bit ADC in this question, if the gain before the ADC increases by 10 decibels, any fixed level at the ADC input corresponds to a receiver input level that is 10 decibels lower. So the input-referred ADC noise floor falls by 10 decibels. The input full-scale level also changes from $-30\,dBm$ to $-40\,dBm$.

If the ADC causes compression, the input one-decibel compression point also falls by about 10 decibels. Full scale, the start of clipping, and the one-decibel compression point are not exactly the same level. Their difference depends on the signal shape.

To explain why the noise figure stops improving, we use

$$F_{total} \simeq F_{front} + \frac{N_{ADC,out}}{GkT_0B}$$

The second term gets smaller as gain increases. Once it is much smaller than the front-end noise, the total noise figure approaches the front-end noise figure. More gain then gives little noise improvement, while the ability to handle strong signals still falls.

3 Design calculations and measurement plan

To choose the noise source, we use the same relation as before:

$$y = 10 \log_{10}\left(1 + 10^{(ENR_{dB} - NF)/10}\right)$$

For a noise figure of about 7 decibels, a 15-decibel ENR source gives a difference of $8.638920\,dB$. For a noise figure of about 30 decibels, a 35-decibel ENR source gives $6.193310\,dB$. In both cases, the linear ratio is far enough from one. We still need to check that the on-state power does not cause compression.

For the averaging time, we use the statistical variation of noise power:

$$\sigma_{P,dB} \simeq \frac{10/\ln 10}{\sqrt{B\tau}}$$

To keep the standard deviation of one reading below $0.05\,dB$ in a 400-kilohertz band, we need

$$\tau > \frac{(4.342945/0.05)^2}{400000} = 0.018861\,s$$

We use at least 0.1 seconds of averaging in each state after a 0.25-second settling time.

For the two-tone test, we must consider the total power. For two equal tones,

$$P_{total} = P_{per\,tone} + 10 \log_{10} 2 = P_{per\,tone} + 3.010300\,dB$$

So the total-power limit of $-10\,dBm$ gives a limit of $-13.010300\,dBm$ per tone. To leave some margin, we choose a practical limit of $-15\,dBm$ per tone. All these powers refer to the receiver input, after the cable, combiner, and attenuator.

For the third-order product to be at least 10 decibels above the noise floor, we need

$$3P_{in} - 2IIP3 \geq N_{floor} + 10$$

or

$$P_{in} \geq \frac{N_{floor} + 10 + 2IIP3}{3}$$

For the model at zero gain, the measured noise figure is about $36.91\,dB$ and $IIP3 = -25\,dBm$. With the 10-kilohertz reference noise floor, the minimum power per tone is about $-45.69\,dBm$. So we choose $-43\,dBm$. This separates the product from the noise floor while staying below ADC clipping.

Using these calculations, we complete the design table as follows.

Quantity	Value	Numerical reason
ENR for $NF \simeq 7$ and $30\,dB$	15 and $35\,dB$	The on/off differences are 8.639 and $6.193\,dB$, respectively.
Attenuation after a 15-decibel ENR source	$NF \simeq 7$: zero; $NF \simeq 30$: not enough	In the second case, Y is only $0.135\,dB$. An attenuator cannot increase ENR.
Attenuation after a 35-decibel ENR source	$NF \simeq 7$: $20\,dB$; $NF \simeq 30$: zero	In the first case, $ENR_{eff} = 35 - 20 = 15\,dB$.
Averaging time	At least $0.1\,s$ after $0.25\,s$ settling	The theoretical time for a $0.05\,dB$ standard deviation in the nominal 400-kilohertz band is $18.9\,ms$.
Single-tone model sweep	$G = 40$: -90 to $-64\,dBm$; $G = 0$: -60 to $-31\,dBm$	The step near compression is one decibel. The hardware limit is $-10\,dBm$ total power.
Tone and product frequencies	$f_1 = 300\,kHz$, $f_2 = 400\,kHz$	The spacing is 100 kilohertz. Products are at 200 and 500 kilohertz. The noise band is centred at $-500\,kHz$.
Power per tone at zero gain	$-43\,dBm$	The product is about 18 decibels above the 10-kilohertz reference floor. The total two-tone power is $-39.99\,dBm$.

The 10-kilohertz reference bandwidth above is exactly the value required by the question.

4 Receiver model simulation

We ran the supplied model in radioconda with GNU Radio 3.10.12.0. This is the supplied 8-bit receiver model. The main model parameters are

$$f_s = 2.048\,MS/s, \qquad NF_{front} = 6\,dB$$
$$IIP3_{front} = -25\,dBm, \qquad P_{FS0} = -30\,dBm$$

Here, P_{FS0} is the input full-scale level at zero gain. We used the original GNU Radio blocks and filters with finite data acquisition. Noise power was averaged in linear units before conversion to decibels. The calibrated model output follows $P_{out}(dBm) = P_{out}(dBFS) - 30$.

4.1

First, we find the noise figure at gains of 40 and 10 decibels. For each case, we took five independent on/off readings and averaged the powers in linear units. The results are below.

$G\ (dB)$	$ENR\ (dB)$	N_{off} (dBFS)	N_{on} (dBFS)	$y\ (dB)$	$NF\ (dB)$
40	15	-41.839	-32.777	9.062	6.513
10	35	-51.391	-42.703	8.688	26.944

For each row, we convert the power ratio back to a linear value and substitute it into the noise figure relation:

$$\begin{aligned} NF_{G=40} &= 15 - 10\log_{10}(10^{9.062173/10} - 1) \\ &\simeq 6.513\,dB \\ NF_{G=10} &= 35 - 10\log_{10}(10^{8.687604/10} - 1) \\ &\simeq 26.944\,dB \end{aligned}$$

So the result is

$$\boxed{NF_{G=40} \simeq 6.51\,dB, \qquad NF_{G=10} \simeq 26.94\,dB}$$

At gain 40, a source with $ENR = 15\,dB$ is suitable. At gain 10, the ADC noise is dominant and the total noise figure is much higher. So we chose $ENR = 35\,dB$ to keep the difference between the readings close to 9 decibels.

4.2

To find the compression point, we first obtain the small-signal output line from a point with enough signal-to-noise ratio and very little distortion. Then we calculate the difference between this line and the output fundamental:

$$C(P) = P + G_{small,FS} - P_{fund,out}(P)$$

We find the input power where $C(P) = 1\,dB$. The sweep results for gains of zero and 40 decibels are shown below.

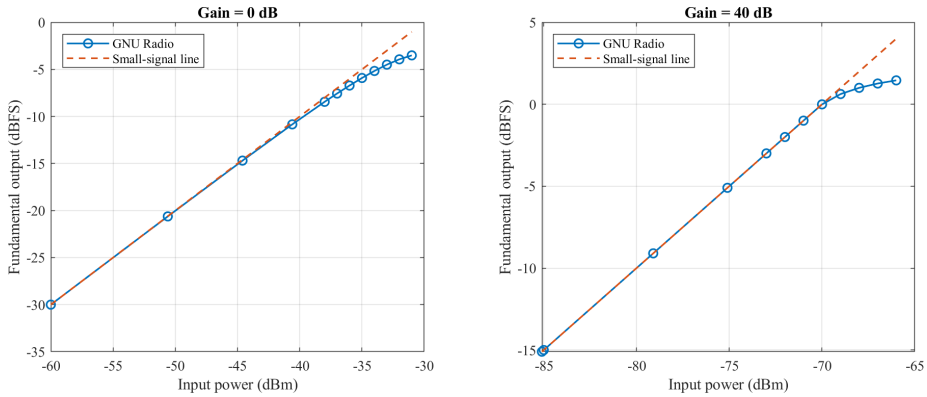


Figure 1: Single-tone sweep, small-signal line, and compression point at gains of zero and 40 decibels

Interpolating between the two points around the compression condition gives

$$\boxed{P_{1dB,G=0} = -34.66\,dBm, \qquad P_{1dB,G=40} = -68.00\,dBm}$$

At zero gain, we can also compare the result with the third-order model:

$$P_{1dB} \simeq -25 - 9.635745 = -34.635745\,dBm$$

This agrees with the sweep result. At this gain, front-end nonlinearity causes compression before the ADC becomes the limit.

At gain 40, a complex tone starts clipping at about $-70\,dBm$. The fundamental falls one decibel below the small-signal line a little later, at about $-68\,dBm$. So the ADC is the limiting stage here.

The sweep step near compression is one decibel; the decimal places come from interpolation.

4.3

At zero gain, we applied two tones at 300 and 400 kilohertz at several input levels. We read the nearby products at 200 and 500 kilohertz. To average the two symmetric components, we first add their linear powers and divide by two, then convert to decibels. Some of the results are below.

P_{in} per tone (dBm)	Fundamental (dBFS)	Raw $IM3$ band (dBFS)	$IM3$ after noise subtraction (dBFS)
-47	-17.166	-60.219	-61.046
-45	-15.265	-54.798	-55.019
-43	-13.423	-48.951	-49.008
-41	-11.681	-42.992	-43.007
-39	-10.105	-36.999	-37.002

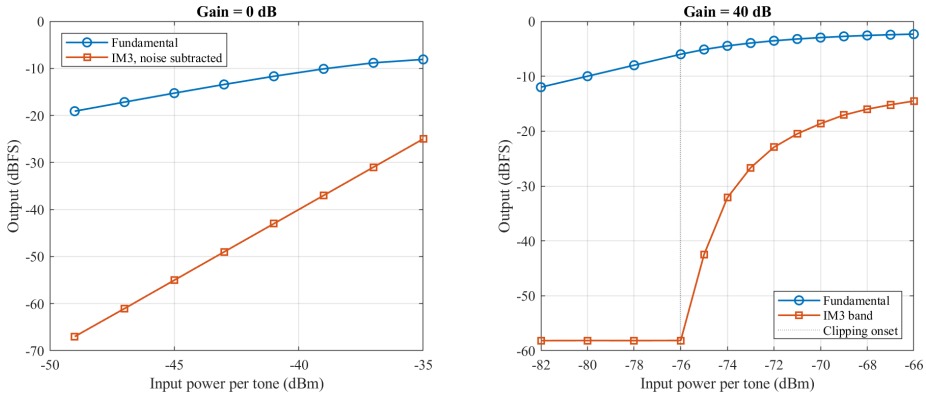


Figure 2: Two-tone test: zero gain on the left and 40-decibel gain on the right

We subtract the separately measured background power in linear units:

$$P_{IM3,net} = P_{band} - P_{noise,band}$$

From -47 to $-43\,dBm$ per tone, the product slope is

$$\boxed{\frac{\Delta P_{IM3,dB}}{\Delta P_{in,dB}} \simeq 3.0095}$$

Using the small-signal fundamental line from the single-tone test gives

$$\boxed{IIP3 \simeq -24.99\,dBm}$$

For comparison,

$$P_{1dB} + 9.6 \simeq -25.06\,dBm$$

The difference is about 0.07 decibels, so the results agree at the sweep resolution.

4.4

Now we repeat the same test at gain 40, shown on the right side of the two-tone figure. Up to about $-76\,dBm$ per tone, the product bands mainly show the noise floor. Above this level, the products appear suddenly.

To explain this level, we consider the sum of the two tone peaks. If the amplitude of each tone at the ADC input is $A_{ADC,tone}$, then at their in-phase peak we have approximately

$$2A_{ADC,tone} \simeq 1$$

So the receiver input power per tone at the start of clipping is approximately

$$\begin{aligned} P_{in,per\,tone} &\simeq P_{FS0} - G - 20\log_{10} 2 \\ &= -76.0206\,dBm \end{aligned}$$

This is the same level where the sudden change appears. For example, when the input goes from -76 to $-75\,dBm$, the product-band reading rises from about -58.15 to $-42.48\,dBFS$. This is a rise of more than 15 decibels for a one-decibel input increase. So the constant slope of three does not hold.

If we put the tone-to-product gap from this region into the $IIP3$ formula, the result changes with the test level. It is not the intercept of a fixed third-order nonlinearity. ADC clipping and higher-order effects are now the limit, so the 9.6-decibel relation also does not apply.

4.5

We tested gains from zero to 50 decibels in steps of 10, as required.

We calculate the reference noise floor in an exact 10-kilohertz band:

$$N_{floor,10kHz} = -173.98 + NF + 10\log_{10}(10^4)$$

The value $-173.98\,dBm/Hz$ follows the model code. In the theory section, we used the rounded value $-174\,dBm/Hz$ given in the question. The gap from the noise floor to compression is

$$D = P_{1dB,in} - N_{floor,10kHz}$$

$G\,(dB)$	$ENR\,(dB)$	$NF\,(dB)$	P_{1dB} (dBm)	N_{floor} (dBm)	$D\,(dB)$
0	45	36.910	-34.656	-97.070	62.414
10	35	26.944	-38.244	-107.036	68.793
20	35	17.290	-48.015	-116.690	68.675
30	15	9.545	-57.998	-124.435	66.437
40	15	6.513	-67.997	-127.467	59.469
50	15	6.055	-78.011	-127.925	49.914

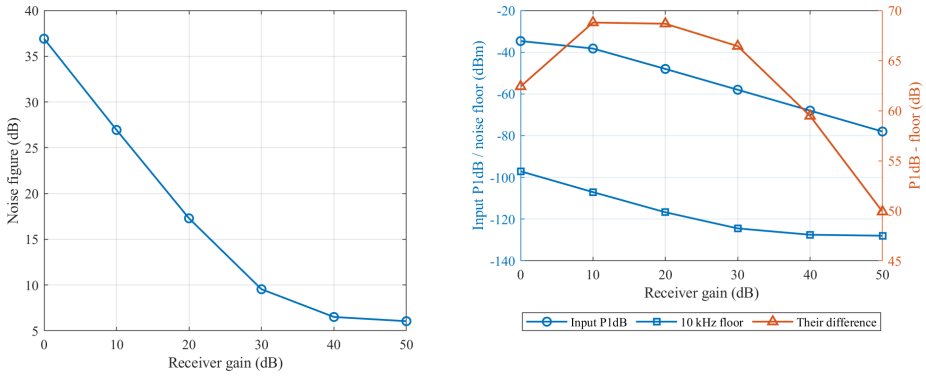


Figure 3: Noise figure and the gap from the noise floor to compression versus gain

Among the requested gain settings, the largest gap is at $G = 10\,dB$:

$$\boxed{D \simeq 68.79\,dB}$$

The reason for the maximum at an intermediate gain is clear. At low gain, ADC noise is dominant, so more gain lowers the input-referred floor. At high gain, the noise figure has already approached the front-end value of about 6 decibels, while the compression point keeps falling.

The gap in the table is based on compression and is different from $SFDR$. To calculate $SFDR$, we need an intercept from a region with a slope of three. This condition does not hold when ADC clipping is the limit.

5 Antenna frequency response simulation

We ran `lab02_antenna_response_simulation.py` and started the 400 to 3000 megahertz sweep. The default frequency step is 20 megahertz. We then used the program's frequency control in one-megahertz steps around the two edges to find the three-decibel points.

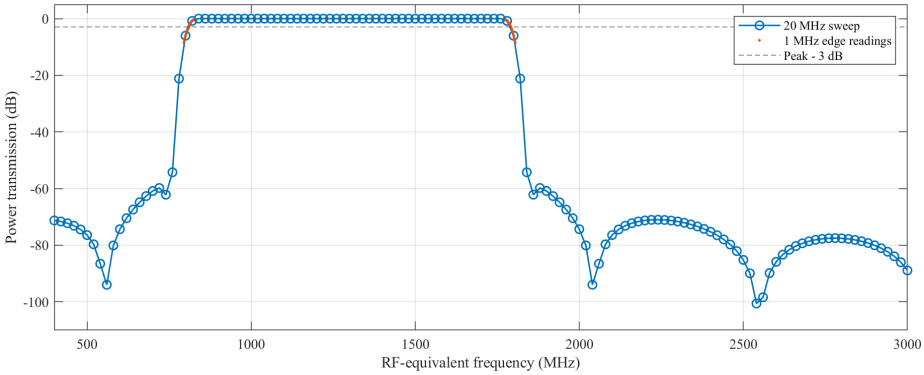


Figure 4: Antenna model response measured with the original program

The largest measured response is about $0.020\,dB$. So the three-decibel level is

$$H_{3dB} = 0.020 - 3 \simeq -2.980\,dB$$

Interpolating between the readings around this level gives

Quantity	Value (MHz)
f_L	808.43
f_H	1791.57
$BW = f_H - f_L$	983.14

Therefore,

$$\boxed{BW = f_H - f_L \simeq 983.14\,MHz}$$

The seller's product table gives minimum and maximum frequencies of 2.4 and 2.5 gigahertz. In megahertz, these are

Band	Lower limit (MHz)	Upper limit (MHz)
1	2400	2500
2	Not stated in the specifications	Not stated in the specifications
3	Not stated in the specifications	Not stated in the specifications

The page also says four bands, but gives no separate frequency limits. The seller's range is not necessarily defined by a three-decibel drop. It is also different from the band of the teaching model. Product specifications.

6 Predictions before the lab session

We use ADALM-PLUTO (PlutoSDR) as the real receiver. It uses the AD9363 and a 12-bit ADC. For these predictions, we assume 800 MHz, room temperature, and manual gain control with AGC off. The AD9363 data sheet lists a typical noise figure of $2.5\,dB$ at 800 MHz and maximum gain. We use about $3\,dB$ as a first estimate for the board, allowing for input loss. This is a prediction, not a measured value. The supplied offline receiver model in Section 4 is still the 8-bit model required by the handout.

If the ADC causes compression, reducing the gain by 10 decibels raises the input compression point by about 10 decibels. If a stage before the variable gain is the limit, this change does not necessarily occur.

For a source with $ENR = 15\,dB$ and a receiver with $NF = 7\,dB$, we have

$$Y = 1 + 10^{(15-7)/10} = 7.309573$$

So the ratio in decibels is

$$\boxed{y = 10\log_{10} Y = 8.638920\,dB}$$

If we place a matched 10-decibel attenuator at the reference temperature before the receiver, the chain noise factor becomes

$$F_{cascade} = L + \frac{F_{rx} - 1}{1/L} = LF_{rx}$$

Therefore,

$$\boxed{NF_{cascade} = NF_{rx} + 10\,dB}$$

This noise figure is referred to the attenuator input. If we remove the attenuator's effect from the measurement, the receiver's own noise figure has not changed.

For a two-decibel increase in each tone, we use the same slopes as before. The fundamental rises by about two decibels, while the third-order product rises by about $3 \times 2 = 6\,dB$. So the gap falls by four decibels. This prediction requires the third-order region to remain valid.

We can now complete the prediction table. The measurement column belongs to the lab session. Our group number is 6. The receiver is ADALM-PLUTO, but the assigned group gain has not been announced.

No.	Quantity	Prediction	Measured
1	Noise figure at maximum gain	About $3\,dB$ at 800 MHz as a first estimate; the AD9363 typical value is $2.5\,dB$ before board input loss	–
2	Noise figure change after a 20-decibel gain reduction	NF generally increases, but not necessarily by 20 decibels. The change depends on which gain stages are reduced	–
3	P_{1dB} change after a 10-decibel gain reduction	About a 10-decibel increase if ADC clipping is the limit; a smaller change is possible if an earlier RF stage limits	–
4	Y for $ENR = 15$ and $NF = 7\,dB$	7.309573 in linear units or $8.638920\,dB$	–
5	Effect of a 10-decibel attenuator before the receiver	A 10-decibel rise in the chain noise figure, for a matched attenuator at the reference temperature	–
6	Product increase when each tone rises by two decibels	About six decibels in the third-order region	–
7	Actual $IIP3 - P_{1dB}$ gap	Not necessarily 9.6 decibels for PlutoSDR. This estimate applies only when cubic RF distortion dominates, not ADC clipping	–
8	Noise figure and compression at group 6's assigned gain	The assigned gain has not been announced	–

7 Likely mistakes in the experiment

7.1

The first mistake is exceeding the receiver input power limit. To prevent this, we calculate the power after the combiner, cable, and attenuator. In the two-tone test, the total power is the power per tone plus $3.0103\,dB$. We turn the source output off when connecting cables and start the sweep at the lowest level. The handout gives a total power limit of $-10\,dBm$. If we use an SDR transmitter, the 30-decibel attenuator required by the handout is also needed.

7.2

The second mistake is measuring the Y ratio at two different gains or while the on state is compressed. To prevent this, we turn off automatic gain control and keep the manual gain and filter fixed. After switching the source state, we allow enough settling and averaging time. We check linearity by repeating the measurement with a lower ENR. The noise figure should stay nearly the same. If Y is close to one, we need a more suitable source level.

7.3

The third mistake is finding a false intercept. This can happen if we read noise as a product, measure distortion from the source, or include ADC clipping. So we measure several input levels and check for a slope of three. Near the noise floor, we also measure the background separately and subtract it in linear units. We check the two-tone source for unwanted products before treating them as receiver distortion.

References

The supplied 11-page guide for Lab 2, titled Noise Figure and RF Parameters, together with the original receiver and antenna model code in `prelab_02.zip`.

Keysight Technologies, application note Noise Figure Measurement Accuracy – The Y-Factor Method, document 5952-3706E, 2014; used for the method definitions and error sources. Application note.

GNU Radio documentation: GNU Radio Wiki.

The antenna specifications were taken from the product table for ANTENNA GSM 3DB 10CM on the Javan Electronic website. Product page.

Analog Devices, ADALM-PLUTO product specifications: AD9363 and 12-bit ADC.

Analog Devices, AD9363 data sheet, Rev. D, Table 1, page 3: typical receiver noise figure at 800 MHz and maximum gain.